Jean Feng

Position

July 2025–Present **Associate Professor**, Department of Epidemiology and Biostatistics, *University of California, San Francisco*

July 2025–Present **Associate Professor**, UCSF-UC Berkeley Joint Program in Computational Precision Health

July 2020–July 2025 **Assistant Professor**, Department of Epidemiology and Biostatistics, *University of California, San Francisco*

Nov 2022–July 2025 **Assistant Professor**, UCSF-UC Berkeley Joint Program in Computational Precision Health

Education

Sept 2015–June 2020 **PhD, Biostatistics**, *University of Washington*, Seattle, WA
Advisors: Noah Simon, Frederick Albert Matsen IV

2012–2013 **MS, Computer Science**, *Stanford University*, Stanford, CA

2009–2013 **BS, Computer Science**, *Stanford University*, Stanford, CA

Publications

Siyu Zhou*, Patrick Vossler*, Venkatesh Sivaraman, Yifan Mai, and Jean Feng. Adaptive auditing of AI systems with anytime-valid guarantees. *Adv. Neural Inf. Process. Syst.*, 2026.

Patrick Vossler, Fan Xia, and Jean Feng. STAVAT: Stratified anytime-valid atom sampling for LLM tier placement. In *The First Conference on Statistics and Trustworthy AI for Cross (X)-Domain Acceleration*, 2026.

Patrick Vossler, Fan Xia, Yifan Mai, Adarsh Subbaswamy, and Jean Feng. LLMs judging LLMs: A simplex perspective. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

Wouter A C van Amsterdam*, Michael Oberst*, Jean Feng*, Jenna Wiens, Shengpu Tang, Shalmali Joshi, Rajesh Ranganath, Mark Sendak, Uri Shalit, Julia E Vogt, Brett Beaulieu-Jones, Muhammad Mamdani, David Kent, Patrick J Heagerty, Thomas R Fleming, and Anna Goldenberg. Clinical trials for continuously monitored and updated AI systems. *Nat. Med.*, 2026.

Jean Feng*, Avni Kothari*, Patrick Vossler, Andrew Bishara, Lucas Zier, Newton Addo, Aaron Kornblith, Yan Shuo Tan, and Chandan Singh. Human-AI co-design for clinical prediction models. *npj Digital Medicine*, 2026.

Venkatesh Sivaraman, Patrick Vossler, Adam Perer, Julian Hong, and Jean Feng. More than "means to an end": Supporting reasoning with transparently designed AI data science processes. *ACM CHI 2026 Workshop on Tools for Thought*, 2026.

Nicholas Phelps, Patrick Vossler, Avni Kothari, Katherine Steineman, Seth Goldman, Arturo Gasga, Susan Ehrlich, Hemal Kanzaria, Julia Adler-Milstein, Jean Feng, Paige Nong, and Lucas S Zier. Patient perspectives about deployment of artificial intelligence decision support tools in a safety-net healthcare system. *JAMIA Open*, 2026.

Kush Narang, Newton Addo, Christopher Y K Williams, Maytal Firnberg, Jean Feng, and Aaron Kornblith. Assessing acuity in pediatric emergency department triage: performance of a large language model. *Npj Health Syst.*, 2026.

Hanyang Li, Xiao He, Adarsh Subbaswamy, Patrick Vossler, Alexej Gossmann, Karandeep Singh, and Jean Feng. Scaling medical device regulatory science using large language models. *NPJ Digit. Med.*, 2026.

Avni Kothari, Patrick Vossler, Jean Digitale, Mohammad Forouzannia, Elise Rosenberg, Michele Lee, Jennee Bryant, Melanie Molina, James Marks, Lucas Zier, and Jean Feng. When the domain expert has no time and the LLM developer has no clinical expertise: Real-world lessons from LLM co-design in a safety-net hospital. *Proc. Conf. AAAI Artif. Intell.*, 2026.

Eileen Kim, Marianna V Elia, Michelle M Li, Elizabeth Healey, Lauren Erdman, Payal Chandak, Peniel Argaw, Ben Li, Kameron C Black, Jayson Marwaha, Hugo Morales, Anna Taylor, Joseph Kvedar, Peter Brodeur, Brett Beaulieu-Jones, Emily Alsentzer, Julian C Hong, and Jean Feng. Clinical AI in 2025–2026 — six themes from the SAIL year in review. *NEJM AI*, 2026.

Chuan Hong, Jonathan Chong Kai Liew, Jaeyong Yu, Tomás Barry, Audrey L Blewer, Daniel M Buckland, Tianrun Cai, Won Chul Cha, Bibhas Chakraborty, Wei Chen, Jun Cheng, Shu-Ling Chong, Therese Djärv, Arul Earnest, Matthew Engelhard, Xiuyi Fan, Mengling Feng, Jean Feng, Huazhu Fu, Wilson Wen Bin Goh, Benjamin A Goldstein, Jessica Gronsbell, Andrew Fu Wah Ho, Kendall Ho, Taku Iwami, Anjni Joiner, Aaron Kornblith, Siqi Li, Shir Lynn Lim, Molei Liu, Zhenghong Liu, Lei Lu, Yuan Luo, Yih Yng Ng, Yilin Ning, Yohei Okada, Ju Ok Park, Yu Rang Park, Junaid Razzak, Yuzeng Shen, Fahad Javaid Siddiqui, Peter A D Steel, Kenneth Boon Kiat Tan, Salinelat Teixayavong, Bella Vakulenko-Lagun, Joao Ricardo Nickenig Vissoci, Grzegorz Waligora, Fei Wang, Haibo Wang, Haoyuan Wang, An-Kwok Ian Wong, Feng Xie, Jie Yang, Yiye Zhang, Doudou Zhou, Li Zhou, Tingting Zhu, Robert Neumar, David Page, Roger Vaughan, Marcus Eng Hock Ong, and Nan Liu. Building a global research network for fair, accountable, interpretable, and responsible AI in emergency care: Protocol for a FAIR-EC study. *JMIR Res. Protoc.*, 2026.

Seyed Mohammad Forouzannia, Jassmine Hempstead, Katherine Steineman, Rong Duan, Patrick Vossler, Jean Feng, Siyu Zhou, Lucas S Zier, and Sammy Elmariah. A proactive digital valve care pathway for management of severe aortic stenosis: The DETECT AS II study. *JAMA Cardiol.*, 2026.

Laleh Foroutani, Allison Chang, Architha Kannan, Amir Ashraf Ganjouei, Jane Wang, Jean Feng, Lucas Thornblade, Kenzo Hirose, Ajay V Maker, Eric Nakakura, Carlos Corvera, Adnan Alseidi, and Mohamed Abdelgadir Adam. Adjusted trend analysis in the adoption and outcomes of minimally invasive pancreaticoduodenectomy approaches: over a decade of contemporary U.S. data. *Surg. Endosc.*, 2026.

Jie Jane Chen, Jean Feng, Isabel D Friesner, Chichi Chang, Clodagh Kenny, Marianna V Elia, Lisa Ni, Felix Y Feng, Michael W Rabow, Steve E Braunstein, Lauren C Boreta, and Julian C Hong. Natural language processing to identify documented pain preceding radiotherapy for bone metastases. *JNCI Cancer Spectrum*, 2026.

Jean Feng, Fan Xia, Karandeep Singh, and Romain Pirracchio. Not all clinical AI monitoring systems are created equal: Review and recommendations. *NEJM AI*, 2025.

Jean Feng, Avni Kothari, Luke Zier, Chandan Singh, and Yan Shuo Tan. Bayesian concept bottleneck models with LLM priors. *Adv. Neural Inf. Process. Syst.*, 2025.

Harvineet Singh, Fan Xia, Alexej Gossmann, Andrew Chuang, Julian C Hong, and Jean Feng. Who experiences large model decay and why? A hierarchical framework for diagnosing heterogeneous performance drift. *International Conference on Machine Learning*, 2025.

Jed Keenan Obra, Chandan Singh, Kenshata Watkins, Jean Feng, Ziad Obermeyer, and Aaron Kornblith. Potential for algorithmic bias in clinical decision instrument development. *NPJ Digit. Med.*, 2025.

Heyuan Michael Ni, Ramez Kouzy, Ali Sabbagh, Michael K Rooney, Jean Feng, Simon P Castillo, Sherif M Gadoue, Zakaria El Kouzi, Karen Hoffman, Yinyin Yuan, Anant Madabhushi, and Osama Mohamad. The state of the art in artificial intelligence and digital pathology in prostate cancer. *Nat. Rev. Urol.*, 2025.

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Alexandre Kalimouttou, Jason N Kennedy, Jean Feng, Harvineet Singh, Suchi Saria, Derek C Angus, Christopher W Seymour, and Romain Pirracchio. Optimal vasopressin initiation in septic shock: The OVISS reinforcement learning study. *JAMA*, 2025.

Laleh Foroutani, Andrew Gonzalez, Jaeyun Jane Wang, Bahaa I Aburayya, Amir Ashraf Ganjouei, Jean Feng, Lucas Willian Thornblade, Kenzo Hirose, Ajay V Maker, Eric Nakakura, Carlos Uriel Corvera, Adnan Alseidi, and Mohamed A Adam. Minimally invasive distal pancreatectomy as the standard of care in the US: Are we there yet? *Cancers (Basel)*, 2025.

Chichi Chang, Jie Jane Chen, Jean Feng, Isabel Friesner, Somya Mohindra, Lauren Boreta, Michael W Rabow, Steve E Braunstein, Ryzen Benson, and Julian C Hong. Patterns in symptoms preceding acute care in patients with cancer. *JAMA Netw. Open*, 2025.

Daniel J Bennett, Jean Feng, Seth Goldman, Avni Kothari, Laura M Gottlieb, Matthew S Durstenfeld, James Marks, Susan Ehrlich, Jonathan Davis, and Lucas S Zier. Reducing readmissions in the safety net through AI and automation. *Am. J. Manag. Care*, 2025.

Jane Wang, Francesca Tozzi, Amir Ashraf Ganjouei, Fernanda Romero-Hernandez, Jean Feng, Lucia Calthorpe, Maria Castro, Greta Davis, Jacquelyn Withers, Connie Zhou, Zaim Chaudhary, Mohamed Adam, Frederik Berrevoet, Adnan Alseidi, and Nikdokht Rashidian. Machine learning improves prediction of postoperative outcomes after gastrointestinal surgery: A systematic review and meta-analysis. *J. Gastrointest. Surg.*, 2024.

Jean Feng, Adarsh Subbaswamy, Alexej Gossmann, Harvineet Singh, Berkman Sahiner, Mi-Ok Kim, Gene Pennello, Nicholas Petrick, Romain Pirracchio, and Fan Xia. Designing monitoring strategies for deployed machine learning algorithms: navigating performativity through a causal lens. *Conference on Causal Learning and Reasoning*, 2024.

Jean Feng, Alexej Gossmann, Romain Pirracchio, Nicholas Petrick, Gene Pennello, and Berkman Sahiner. Is this model reliable for everyone? testing for strong calibration. *AISTATS*, 2024.

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Jean Feng, Adarsh Subbaswamy, Alexej Gossmann, Harvineet Singh, Berkman Sahiner, Mi-Ok Kim, Gene Pennello, Nicholas Petrick, Romain Pirracchio, and Fan Xia. Towards a Post-Market monitoring framework for machine learning-based medical devices: A case study. *Workshop on Regulatable Machine Learning at the 37th Conference on Neural Information Processing Systems*, 2023.

Alexandra B Steverson, Paul J Marano, Caren Chen, Yifei Ma, Rachel J Stern, Jean Feng, Efstathios D Gennatas, James D Marks, Matthew S Durstenfeld, Jonathan D Davis, Priscilla Y Hsue, and Lucas S Zier. Predictors of All-Cause Thirty-Day readmissions in patients with heart failure at an urban safety net hospital: The importance of social determinants of health. *American Journal of Medicine Open*, 2023.

Harvineet Singh, Fan Xia, Mi-Ok Kim, Romain Pirracchio, Rumi Chunara, and Jean Feng. A brief tutorial on sample size calculations for fairness audits. *Workshop on Regulatable Machine Learning at the 37th Conference on Neural Information Processing Systems*, 2023.

Ali Sabbagh, Samuel L Washington, 3rd, Derya Tilki, Julian C Hong, Jean Feng, Gilmer Valdes, Ming-Hui Chen, Jing Wu, Hartwig Huland, Markus Graefen, Thomas Wiegel, Dirk Bohmer, Janet E Cowan, Matthew Cooperberg, Felix Y Feng, Mack Roach, 3rd, Bruce J Trock, Alan W Partin, Anthony V D'Amico, Peter R Carroll, and Osama Mohamad. Development and external validation of a machine learning model for prediction of lymph node metastasis in patients with prostate cancer. *Eur Urol Oncol*, 2023.

Ali Sabbagh, Derya Tilki, Jean Feng, Hartwig Huland, Markus Graefen, Thomas Wiegel, Dirk Böhmer, Julian C Hong, Gilmer Valdes, Janet E Cowan, Matthew Cooperberg, Felix Y Feng, Tarek Mohammad, Mohamed Shelan, Anthony V D'Amico, Peter R Carroll, and Osama Mohamad. Multi-institutional development and external validation of a machine learning model for the prediction of distant metastasis in patients treated by salvage radiotherapy for biochemical failure after radical prostatectomy. *European Urology Focus*, 2023.

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Jackie J Lin, Patricia C Conroy, Fernanda Romero-Hernandez, Mignote Yilma, Jean Feng, Kenzo Hirose, Eric Nakakura, Ajay V Maker, Carlos Corvera, Kimberly Kirkwood, Adnan Alseidi, and Mohamed A Adam. Hypertension requiring medication use: a silent predictor of poor outcomes after pancreaticoduodenectomy. *J. Gastrointest. Surg.*, 2023.

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Jean Feng and Noah Simon. Ensembled sparse-input hierarchical networks for high-dimensional datasets. *Statistical Analysis and Data Mining*, 2022.

Jean Feng, Rachael V Phillips, Ivana Malenica, Andrew Bishara, Alan E Hubbard, Leo A Celi, and Romain Pirracchio. Clinical artificial intelligence quality improvement: towards continual monitoring and updating of AI algorithms in healthcare. *npj Digital Medicine*, 2022.

Jean Feng, Gene Pennello, Nicholas Petrick, Berkman Sahiner, Romain Pirracchio, and Alexej Gossmann. Sequential algorithmic modification with test data reuse. *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, 2022.

Jean Feng, Alexej Gossmann, Berkman Sahiner, and Romain Pirracchio. Bayesian logistic regression for online recalibration and revision of risk prediction models with performance guarantees. *Journal of the American Medical Informatics Association*, 2022.

Daniel Lazzareschi, Ravindra L Mehta, Laura M Dember, Juliane Bernholz, Alparslan Turan, Amit Sharma, Sachin Kheterpal, Chirag R Parikh, Omar Ali, Ivonne H Schulman, Abigail Ryan, Jean Feng, Noah Simon, Romain Pirracchio, Patrick Rossignol, and Matthieu Legrand. Overcoming barriers in the design and implementation of clinical trials for acute kidney injury: a report from the 2020 kidney disease clinical trialists meeting. *Nephrol. Dial. Transplant*, 2022.

Andre Esteva, Jean Feng, Douwe van der Wal, Shih-Cheng Huang, Jeffry P Simko, Sandy DeVries, Emmalyn Chen, Edward M Schaeffer, Todd M Morgan, Yilun Sun, Amirata Ghorbani, Nikhil Naik, Dhruv Nathawani, Richard Socher, Jeff M Michalski, Mack Roach, Thomas M Pisansky, Jedidiah M Monson, Farah Naz, James Wallace, Michelle J Ferguson, Jean-Paul Bahary, James Zou, Matthew Lungren, Serena Yeung, Ashley E Ross, Howard M Sandler, Phuoc T Tran, Daniel E Spratt, Stephanie Pugh, Felix Y Feng, and Osama Mohamad. Prostate cancer therapy personalization via multi-modal deep learning on randomized phase III clinical trials. *npj Digital Medicine*, 2022.

Jean Feng, Arjun Sondhi, Jessica Perry, and Noah Simon. Selective prediction-set models with coverage guarantees. *Biometrics*, 2021.

Jean Feng, Scott Emerson, and Noah Simon. Approval policies for modifications to Machine Learning-Based software as a medical device: A study of bio-creep. *Biometrics*, 2021.

Jean Feng, William S DeWitt, Aaron McKenna, Noah Simon, Amy Willis, and Frederick A Matsen. Estimation of cell lineage trees by maximum-likelihood phylogenetics. *Annals of Applied Statistics*, 2021.

Jean Feng. Learning to safely approve updates to machine learning algorithms. *Proceedings of the Conference on Health, Inference, and Learning*, 2021.

Brian D Williamson and Jean Feng. Efficient nonparametric statistical inference on population feature importance using shapley values. *International Conference on Machine Learning*, 2020.

Jean Feng and Noah Simon. An analysis of the cost of hyper-parameter selection via split-sample validation, with applications to penalized regression. *Statistica Sinica*, 2020.

Jean Feng, David A Shaw, Vladimir N Minin, Noah Simon, and Frederick A Matsen, IV. Survival analysis of DNA mutation motifs with penalized proportional hazards. *Ann. Appl. Stat.*, 2019.

Kristian Davidsen, Branden J Olson, William S DeWitt, 3rd, Jean Feng, Elias Harkins, Philip Bradley, and Frederick A Matsen, 4th. Deep generative models for T cell receptor protein sequences. *Elife*, 2019.

Jean Feng, Brian Williamson, Noah Simon, and Marco Carone. Nonparametric variable importance using an augmented neural network with multi-task learning. *International Conference on Machine Learning*, 2018.

Jean Feng and Noah Simon. Gradient-based regularization parameter selection for problems with nonsmooth penalty functions. *J. Comput. Graph. Stat.*, 2018.

Pre-prints

Siyu Zhou, Jialin Ouyang, Fan Xia, Adarsh Subbaswamy, Drew Prinster, Suchi Saria, Augusto Garcia-Agundez, Sarah Pollet, Jinoos Yazdany, Patrick Vossler, and Jean Feng. Learning to detect shifts in unstructured data streams via monitoring-by-betting. *arXiv*, 2026.

Patrick Vossler*, Jean Feng*, Venkat Sivaraman, Robert Gallo, Hemal Kanzaria, Dana Freiser, Christopher Ross, Amy Ou, James Marks, Susan Ehrlich, Christopher Peabody, and Lucas Zier. From fuzzy to formal: Scaling hospital quality improvement with AI. *arXiv*, 2026.

Patrick Vossler, Jialin Ouyang, Richard Guo, Anran Huang, Ali Shojaie, Lucas Zier, Fan Xia, and Jean Feng. Expert-guided g-computation with large language models for estimating causal effects on timings: Applications to hospital quality improvement. *arXiv*, 2026.

Jean Feng, Vishal Patel, Patrick Heagerty, Yifan Mai, Venkatesh Sivaraman, Patrick Vossler, Jialin Ouyang, and Anupam B Jena. Expert evaluation of clinical AI tools on real point-of-care clinical queries. *arXiv*, 2026.

Venkatesh Sivaraman, Rigney Turnham, George Bonano, Nevin Aresh, Renumathy Dhanasekaran, Margaret Guo, Sindhu Kubendran, Olivia Lin, Jonathan D Louie, Kristan Olazo, Jeanne Shen, Harish Vasudevan, Jeanette Wong, Emily Alsentzer, Jason A Fries, Anobel Odisho, John Gordan, Jean Feng, and Julian C Hong. "I know where to look," but does the LLM? charting the gaps between clinical expert needs and unstructured data abstraction tools. *arXiv*, 2026.

Venkatesh Sivaraman, Joel Levin, Eric P Mason, Victor Talisa, Suchi Saria, Jean Feng, Andrew J King, Jeremy M Kahn, and Adam Perer. Indirect influences of artificial intelligence advice on complex sepsis treatment decisions. *arXiv*, 2026.

Kush Narang, Newton Addo, Jaskaran Bains, Maytal Firnberg, Sonny Tat, Anneka Hooft, Daniela Chanci Arrubla, Baraa Abed, Chandan Singh, Rishikesan Kamaleswaran, Jean Feng, and Aaron Kornblith. Learning the electronic health record at the minute-scale. *medRxiv*, 2026.

Augusto Garcia-Agundez, Siyu Zhou, Jessica Pourian, Catherine Blebea, Parnaz Daneshpajouhnejad, Elizabeth Dente, Kevin Shi, Sarah Pollet, Fan Xia, Xu Shi, Robert Thombley, Cynthia Fenton, Sara G Murray, Julia Adler-Milstein, Gabriela Schmajuk, Jean Feng, and Jinoos Yazdany. A framework to monitor editing of artificial intelligence-generated medical documentation. *medRxiv*, 2026.

Will Zou, Jean Feng, Alexandre Kalimouttou, Jennifer Zhang, Christopher Seymour, and Romain Pirracchio. Realistic CDSS drug dosing with end-to-end recurrent Q-learning for dual vasopressor control. *NeurIPS 2025 Workshop on Learning from Time Series for Health*, 2025.

Jean Feng and Noah Simon. Sparse-Input neural networks for high-dimensional nonparametric regression and classification. *arXiv*, 2019.

Abstracts

A Sabbagh, D Tilki, J Feng, J C Hong, M H Chen, J Wu, H Huland, M Graefen, T Wiegel, D Böhmer, S Washington, III, J Cowan, M R Cooperberg, F Y Feng, P Carroll, B Trock, A W Partin, A V DAmico, and O Mohamad. Machine learning for the prediction of distant metastases following postprostatectomy salvage radiation therapy. *Int. J. Radiat. Oncol. Biol. Phys.*, 2022.

A Sabbagh, S Washington Iii, D Tilki, J C Hong, J Feng, M H Chen, J Wu, H Huland, M Graefen, T Wiegel, D Böhmer, J Cowan, M R Cooperberg, F Y Feng, B Trock, A W Partin, A V DAmico, P Carroll, and O Mohamad. Machine learning for the prediction of lymph node metastasis in patients with prostate cancer. *Int. J. Radiat. Oncol. Biol. Phys.*, 2022.

I Friesner, J Feng, S Kalnicki, M K Garg, N Ohri, and J C Hong. Machine Learning-Based prediction of hospitalization using daily step counts for patients undergoing chemoradiation. *Int. J. Radiat. Oncol. Biol. Phys.*, 2022.

Denise Cecil, Jean Feng, Alex Paynter, Jessica Perry, Noah Simon, Nicholas Drovetto, Lauren Corulli, Erin Rodmaker, Susan Strenk, David Fredricks, and Mary Disis. 1308 bacteria specific IL-10 secreting t-cells derived from the gut are cross-reactive with tumor antigens and accelerate tumor growth in mouse models. *J Immunother Cancer*, 2022.

Presentations

Invited Oral Presentations and Seminars

2026 *The Role of AI in Supporting the Full Lifecycle of Hospital Quality Improvement*, UCSD Health AI Grand Rounds

2026 *Advancing Methods of AI in Comparative Clinical Effectiveness Research*, PCORI Annual Conference

2026 *Clinical AI: Year in Review*, Boston Children's Hospital, Computational Health Informatics Seminar Series

2026 *Bayesian Concept Bottleneck Models with LLM Priors*, Joint Statistical Meetings (JSM)

2026 *Bayesian Concept Bottleneck Models with LLM Priors*, International Biometrics Conference

2026 *AI at an Urban Safety-net Hospital*, ACM Conference on Health, Inference, and Learning (CHIL)

2026 *LLMs Judging LLMs and Humans Judging LLMs: Who Can We Trust?*, Fred Hutchinson Cancer Research Center, Biostatistics Seminar Series

2026 *Clinical AI: Year in Review*, Symposium on Artificial Intelligence in Learning Health Systems (SAIL)

2026 *Clinical Reasoning in the Era of Artificial Intelligence (Panel)*, UC Health - Center for Data-driven Insights and Innovation (CDI2) Conference

2026 *Human-AI Co-design for Clinical Prediction Models*, Centers for Disease Control and Prevention (CDC) AI Community of Practice

2026 *Bayesian Concept Bottleneck Models with LLM Priors*, Online Interpretable Deep Learning Reading Group

2026 *Towards a Post-Market monitoring framework for AI/ML-based medical devices*, Forum on the Integration of Observational and Randomized Data (FIORD)

2026 *Bayesian Concept Bottleneck Models with LLM Priors*, Eastern North American Region (ENAR) Meeting

2026 *Toward statistically rigorous LLM-powered analyses: Two case studies*, University of California, Riverside, Department of Statistics Seminar

2026 *Monitoring clinical AI systems: Where are we now and where do we want to be?*, Simons Institute for the Theory of Computating, Workshop on Bridging Prediction and Intervention Problems in Social Systems

2026 *Scaling Medical Device Regulatory Science using Large Language Models*, Center of Excellence in Regulatory Science and Innovation (CERSI) Summit

2025 *LLMs judging LLMs: A Simplex Perspective*, Peking University, Biostatistics Seminar

2025 *Beyond the benchmark dataset: Real-world generalizability and regulatory challenges in medical AI*, Medical Image Computing and Computer Assisted Intervention Society (MICCAI) Special Interest Group for Biomedical Image Analysis Challenges

2025 *Monitoring clinical AI: Where are we now and where do we want to be?*, Machine Learning for Healthcare (MLHC) pre-conference workshop

2025 *External Validation of Clinical AI Models: Addressing Performance Decay and Distribution Shifts*, Symposium for the Launch of the Center for External AI validation in Healthcare, Taiwan Ministry of Health and Welfare

2025 *Translational Research in a Rapidly Changing World*, ACM Conference on Health, Inference, and Learning

2025 *Opportunities to leverage LLMs across the clinical AI/ML lifecycle*, Deep Learning and Health Futures Seminar, Microsoft Research

2025 *Are we ready to monitor clinical ML/AI algorithms?*, Biostatistics Seminar Series, University of Washington

2025 *Bayesian Concept Bottleneck Models with LLM Priors*, Deep Learning Affinity Group, Fred Hutchinson Cancer Center

2025 *Bayesian Concept Bottleneck Models with LLM Priors*, University of Southern California, SEEDS (Statistics Empowering Data Science) Conference

2025 *Towards a Post-Market monitoring framework for machine learning-based medical devices*, International Conference on Health Policy Statistics (ICHPS)

2024 *Bayesian Concept Bottleneck Models with LLM Priors*, FDA OSEL AI/ML Program Research Update Meeting

2024 *Strategies for Developing and Implementing AI/ML in Underserved Healthcare Environments*, AMIA Anual Symposium

2024 *Towards a Post-Market monitoring framework for machine learning-based medical devices*, Joint seminar between UC Berkeley Center for Targeted Machine Learning and the UCSF-UC Berkeley Computational Precision Health Program

2024 *Towards a Post-Market monitoring framework for machine learning-based medical devices*, Brown University, Symposium celebrating 30 Years of Biostatistics at Brown University

2024 *Bayesian Priors From Large Language Models Make Clinical Prediction Models More Interpretable*, AMIA 2024 Annual Symposium

2024 *"Why did the AUC drop?" A Hierarchical Framework to Explain Performance Changes of Machine Learning Models across Hospital Sites*, AMIA 2024 Annual Symposium

2024 *Towards a Post-Market monitoring framework for machine learning-based medical devices*, Johnson and Johnson MedTech Data Science Seminar

2024 *Designing monitoring strategies for deployed machine learning algorithms: navigating performativity through a causal lens*, Joint Statistical Meetings

2024 *Is this model reliable for everyone? Testing for strong calibration*, International Conference on Econometrics and Statistics

2024 *The Science Behind Monitoring and Updating AI Models*, Symposium on Artificial Intelligence for Learning Health Systems (SAIL)

2024 *Quality assurance and improvement for Machine Learning-based medical devices*, AI Regulatory and International Symposium (AIRIS)

2024 *Towards a Post-Market monitoring framework for machine learning-based medical devices: A case study*, CERSI Scientific Symposium

2023 *Statistical Tools for Auditing Machine Learning Algorithms Across Subgroups and Time*, IMS Young Mathematical Scientists Forum

2023 *Statistical Tools for Auditing Machine Learning Algorithms Across Subgroups and Time*, Bay Area Biotech-Pharma Statistics Workshop

2023 *Sequential algorithmic modification with test data reuse*, Medical Imaging and Data Resource Center Seminar

2023 *Is this risk prediction model reliable for everyone? A test for strong model calibration*, Joint Statistical Meetings

2023 *Monitoring machine learning (ML)-based risk prediction algorithms in the presence of confounding medical interventions*, Joint Statistical Meetings

2023 *Monitoring machine learning (ML)-based risk prediction algorithms in the presence of confounding medical interventions*, International Chinese Statistical Association Applied Statistics Symposium

2023 *Quality assurance and improvement for Machine Learning-based clinical decision support systems*, UC Berkeley Biostatistics Seminar

2023 *Monitoring machine learning (ML)-based prediction algorithms in the presence of confounding medical interventions*, AMIA 2023 Informatics Summit

2022 *Efficient nonparametric statistical inference for population variable importance*, IMS International Conference on Statistics and Data Science (ICSDS)

2022 *Opportunities at the intersection of Machine Learning and Epidemiology*, Society for Epidemiologic Research (SER) Digital

2022 *Quality assurance and improvement for Machine Learning-based clinical decision support systems*, Colorado School of Public Health

2022 *Statistical Methods for Monitoring and Updating AI/ML-Based Software as a Medical Device*, ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop

2022 *Sequential algorithmic modification with test data reuse*, Uncertainty in Artificial Intelligence

2022 *Approval policies for modifications to Machine Learning-Based Software as a Medical Device: A study of bio-creep*, International Biometrics Society Journal Club

2021 *Safe approval policies for continual learning systems in healthcare*, Brown University

2021 *Bayesian logistic regression for online recalibration and revision of risk prediction models with guarantees*, ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop

2021 *Variable Selection and Architecture Search for Neural Networks*, ASA Statistical Learning and Data Science Webinar

2021 *Bayesian logistic regression for online recalibration and revision of risk prediction models with guarantees*, Western North American Region (WNAR) Annual Meeting

2021 *Learning to safely approve updates to machine learning algorithms*, ACM Conference on Health, Inference, and Learning

2021 *Safe approval policies for continual learning systems in healthcare*, University of Waterloo

2020 *Efficient nonparametric statistical inference on population feature importance using Shapley values*, International Conference on Machine Learning

2020 *Training Procedures and Regulatory Policies for Safe Machine Learning Models in Healthcare*, University of California, San Francisco

2020 *Training Procedures and Regulatory Policies for Safe Machine Learning Models in Healthcare*, The University of Texas, MD Anderson Cancer Center

2020 *Training Procedures and Regulatory Policies for Safe Machine Learning Models in Healthcare*, University of California, Irvine

2020 *Approval policies for modifications to Machine Learning-Based Software as a Medical Device: A study of Bio-creep*, International Conference on Health Policy Statistics

2019 *Sparse-Input Neural Networks for High-dimensional Nonparametric Regression and Classification*, Western North American Region (WNAR) Annual Meeting

2018 *Nonparametric variable importance using an augmented neural network with multi-task learning*, International Conference on Machine Learning

2018 *Sparse-Input Neural Networks for High-dimensional Nonparametric Regression and Classification*, University of Washington Biostatistics Colloquium

2018 *Sparse-Input Neural Networks for High-dimensional Nonparametric Regression and Classification*, Joint Statistical Meetings

2017 *Sparse-Input Neural Networks for High-dimensional Nonparametric Regression*, ICML Workshop on Principled Approaches to Deep Learning

2011 *Haptic Belt with Pedestrian Detection*, Neural Information Processing Systems

Contributed Oral Presentations

2025 *Accelerating 510(k) Predicate Device Selection Using Large Language Models and Data Visualization*, Innovations in Regulatory Science Summit

2023 *Monitoring machine learning (ML)-based risk prediction algorithms: Addressing the challenge of confounding medical interventions*, 2023 Innovations in Regulatory Science Summit

2019 *Uncertainty-Aware Black-Box Predictors with Coverage Guarantees*, Joint Statistical Meetings

Funding History

2026-2027 US Food and Drug Administration Broad Agency Announcement (BAA); Role: PI
"Scalable monitoring procedures for generative AI-based medical devices": $730,000 total

2026-2027 UCSF Department of Epidemiology and Biostatistics, Artificial Intelligence Extramural Collaboration Seed Grant; Role: PI
"With AI Agents, You Too Can Build Interpretable Prediction Models from Unstructured Data: *An Open-Source Software Package*": $25,000 in direct costs

2026-2027 UCSF Department of Epidemiology and Biostatistics, Artificial Intelligence/Machine Learning for Health Seed Grant; Role: co-PI
"Post-market surveillance of AI Scribes with LLM-based causal monitoring": $50,000 in direct costs

2023-2026 Patient-Centered Outcomes Research Institute; Role: PI
"Diagnostic tools for quality improvement of machine learning-based clinical decision support systems": $750,000 in direct costs
+ "Methods Supplement to Support Innovative Research on AI and Large Language Models in Patient-Centered CER": $225,000 in direct costs

2020-2023 UCSF-Stanford CERSI Program; Role: PI
"Safe algorithmic change protocols for modifications to AI/ML-based Software as a Medical Device": $170,000 in direct costs

Awards

2026 Paper Award, Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X)

2026 Stellar Abstract Award, Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X)

2026 Service Award, Symposium on Artificial Intelligence in Learning Health Systems (SAIL)

2024 Quality Leaders Award in Equity, California Association of Public Hospitals and Health Systems Safety Net Institute

2024 The Joint Commission's Bernard J. Tyson National Award for Pursuit of Healthcare Equity

2020 University of Washington Thomas R. Fleming Excellence in Biostatistics Award

2020 International Conference on Health Policy Statistics, Student Travel Award
For manuscript: *Approval policies for modifications to Machine Learning-Based Software as a Medical Device*

2018 Joint Statistical Meetings Section on Statistical Learning and Data Science, Student Paper Award
For manuscript: *Sparse-input neural networks for high-dimensional nonparametric regression and classification*

2018 University of Washington Biostatistics Donovan J. Thompson Award for Best Combined Performance on Ph.D. Theory and Applied Qualifying Examinations

2015–2017 Big Data for Genomics and Neuroscience Training Grant

Software

EASIER-Net Python and R packages for fitting neural networks for high-dimensional data
`https://github.com/jjfeng/easier_net`
`https://github.com/jjfeng/easier_net_R`

GapML Python package for analyzing cell-lineage tracing data from GESTALT
`https://github.com/matsengrp/gestaltamania`

SPINN Python package for estimating sparse-input neural networks
`http://github.com/jjfeng/spinn`

samm Python package for estimating somatic hypermutation rates of nucleotide motifs
`http://github.com/matsengrp/samm`

Teaching

2024-Present Instructor, Datsci 224: Understanding Machine Learning: From Theory to Applications, UCSF

2026-Present Instructor, Summer Institute in Statistics for Big Data (SISBID): Deep Learning and Artificial Intelligence, University of Washington

2024-2025 Instructor, Summer Institute in Statistics for Big Data (SISBID): Supervised Methods for Statistical Machine Learning, University of Washington

2025 Guest Lecture, Taiwan Smart Healthcare School, Department of Information Management, Taiwan Ministry of Health and Welfare

2021-2023 Instructor, Biostat 216: Machine Learning in R for the Biomedical Sciences, UCSF

2022 Lecturer, Epidemiology, Biostatistics and Population Science (EBPS), UCSF School of Medicine Bridges Curriculum

2020-Present Instructor, Machine Learning Boot Camp: Analyzing Biomedical and Health Data, Columbia University

2020 Instructor, Supervised statistical learning, 6th Seattle Symposium in Biostatistics

Student Advising

2026-Present Venkatesh Sivaraman, Postdoctoral researcher

2025-Present Siyu Zhou, Postdoctoral researcher

2026-Present Noah Baker, PhD Biological and Medical Informatics, Committee Member

2025-Present Oscar Peng, Master's student in UCSF Clinical and Epidemiological Research

2023-Present Jinhee Lee, UCSF-UC Berkeley Computational Precision Health Designated Emphasis program, UCSF-UC Berkeley Bioengineering PhD program

2023-Present Saigeetha Bhaskar, Master's student in UCSF Clinical and Epidemiological Research

2023-2025 Harvineet Singh, Postdoctoral researcher

Summer 2024 Shreya Nadagowda, High School Intern from UCSF Summer Researchers in Global Health

Spring 2022 Margaret Tsui, Biomedical Informatics PhD program, Qualifying Exam Committee

Summer 2022 Amanda Everitt, Biomedical Informatics PhD program, Qualifying Exam Committee

2022-2025 Richard Hu, Master's student in UCSF Clinical and Epidemiological Research

2022-2023 Devika Krishnakumar, Master's student in UCSF Clinical and Epidemiological Research

Faculty Advising

2023-Present Andrew Bishara, Assistant Professor, K23 grant

2026-Present Fan Xia, Assistant Professor, Department of Epidemiology and Biostatistics Artificial Intelligence/Machine Learning for Health Seed Grant

Service

Editor

2026-Present Area Chair, International Conference on Machine Learning (ICML)

2026-Present Guest Editor of Special Collection on "Statistical Foundations of AI and Real-World Evidence Generation", Biostatistics

2026-Present Area Chair and Paper Awards Committee, Statistics and Trustworthy AI for Cross-Domain Acceleration (STAI-X)

2026-Present Program Committee, Symposium on Artificial Intelligence in Learning Health Systems (SAIL)

2025-Present	Area Chair, Conference on Neural Information Processing Systems (NeurIPS)
2024-Present	Statistical Editor, New England Journal of Medicine Artificial Intelligence (NEJM AI)
2023-Present	Area Chair, Conference on Health, Inference, and Learning (CHIL)
2023-2025	Editor of Special Issue on "AI/ML Use Cases in Medical Product Development", Journal of Biopharmaceutical Statistics

Referee Service

- BMJ Open Quality
- NEJM AI
- Journal of Machine Learning Research
- Conference on Neural Information Processing Systems
- NeurIPS Workshop on Distribution Shifts
- Frontiers in Digital Health
- Lancet Digital Health
- Nature Digital Medicine
- Nature Medicine
- International Conference on Machine Learning
- International Conference on Learning Representations
- Journal of Computational and Graphical Statistics
- Statistics in Biopharmaceutical Research
- Biometrics
- Annals of Applied Statistics
- Annals of Statistics
- Statistics in Medicine
- Neural Networks
- ASA Section on Statistical Learning and Data Science Student Paper Committee
- Communications Medicine
- JAMA
- Journal of the American Medical Informatics Association (JAMIA)
- Nature Biomedical Engineering
- IEEE Transactions on Neural Systems & Rehabilitation Engineering

Conferences

- Joint Statistical Meetings, Session Chair
- Western North American Region (WNAR) Annual Meeting, Session Chair
- Eastern North American Region (ENAR) Annual Meeting, Roundtable leader

University Service

- UCSF, Department of Epidemiology and Biostatistics, Faculty Search Committee, Fall 2025
- UCSF Women in Tech for International Women's Day, March 2024
- UCSF "Implementation and Evaluation of AI in Real-World Clinical Settings" Seminar Series, Jan 2024–Present
- UCSF Artificial Intelligence Oversight Committee, March 2023–Present
- UCSF Artificial Intelligence/Machine Learning Demonstration Projects Steering Committee, Winter-Spring 2022
- UCSF Initiative for Digital Transformation in Computational Biology & Health Grant Review Committee, September 2021

- Department of Epidemiology and Biostatistics, Digital Health Initiative Steering Committee, 2020–Present
- Department of Epidemiology and Biostatistics, Admissions Committee for the Master's Degree in Health Data Science, 2022–Present
- University of Washington, School of Public Health AI Task Force, Spring 2026

Work Experience

2019 **Research Intern**, *Insitro*, South San Francisco, CA
Developed statistical models of genomic data.

2012–2015 **Software engineer**, *Coursera*, Mountain View, CA
Built the professional certificate program and payment system. Technical lead on projects with 3-5 people. Mentored interns and junior engineers.

Other

Winter 2025 Guest speaker for Monta Vista Robotics Team Diversity in STEM Symposium

Summer 2023 Guest speaker for Kode with Klossy Summer program

2022-present Guest lecturer for UCSF AI4ALL Summer program